

Empowering Near-URLLC IoT with 5G DECT-2020 NR: Current State and the Road Ahead

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ABSTRACT

DECT-2020 is a standardized multi-hop technology for 5G massive machine-type communications (mMTC). By relying on multi-hop communications, DECT-2020 allows for cost-controlled flexible deployments that enable a plethora of new services, whose requirements in terms of latency, reliability, and throughput are difficult to satisfy by utilizing cellular Internet-of-Things technologies (CIoT). In this article, we outline new DECT-2020 use-cases ranging from mMTC to ultra-reliable low-latency communications (URLLC), showing that many modern services lie between these two extremes. We utilize system level simulations to demonstrate how these “near-URLLC” use-cases are satisfied by the DECT-2020. We show that the current evolution of DECT-2020 can provide near-URLLC operation when no multi-hop communications are required, even under interference from competing URLLC/mMTC traffic. However, provisioning of near-URLLC services over multiple hops requires additional enhancements.

INTRODUCTION

The main services specified by ITU-R and 3GPP for 5G systems are enhanced mobile broadband (eMBB), massive machine type communications (mMTC), and ultra-reliable low-latency communications (URLLC). The eMBB service has already been supported by the current evolution of 3GPP 5G New Radio (NR) systems. The mMTC is expected to be delivered via cellular Internet of Things (CIoT) solutions such as LTE-M enhancements for machine type communications (eMTC) and Narrowband Internet-of-Things (NB-IoT), as well as unlicensed band solutions, such as DECT-2020. However, the support for URLLC IoT services lags behind.

DECT-2020 NR is a 5G technology developed for mMTC and URLLC operating in the license-exempt and licensed bands below 6 GHz in local self-hosted deployments [1]. Recent studies [2, 3] demonstrated that it can support up to multiple millions of radio devices (RDs) per square kilometer while satisfying the performance guarantees

for 5G mMTC applications outlined in [4, 5] of no more than 1 percent packet loss rate (PLR) and ensuring that 99 percent of packets experience end-to-end latency smaller than 10 seconds.

The modern IoT market is characterized by the appearance of new applications that fall between URLLC and mMTC. Use-cases such as industrial automation on the factory floor, connectivity between unmanned aerial vehicles (UAV), energy grid monitoring are all characterized by stricter latency, higher reliability, and higher throughput requirements than those specified for mMTC [4]. While they are still not as strict as those defined for URLLC, CIoT solutions such as LTE-M and NB-IoT do not meet the throughput requirements of these use-cases due to limited bandwidth and may fail to deliver latency and reliability requirements.

DECT-2020 operates over larger bandwidth than CIoT solutions, utilizes reliable link-layer mechanisms and simple access regime, and is capable of operating in either single- or multi-hop topologies. Differently from other customer-owned solutions, DECT-2020 NR is a standardized one, is not limited to mMTC service, enables operation in the non-regulated DECT bands, thus, allowing for flexible configurations and network control. It also induces flexibility in radio part implementation by allowing for different levels of implementations depending on the customer needs. These characteristics allow to consider it as one of the solutions for near-URLLC use-cases. The aim of this article is to report the results of the Ultra-Scalable Wireless Access (USWA) Project [6], where we characterized the performance requirements of these use-cases, benchmarked their performance using the DECT-2020 system, and outlined the enhancements that need to be introduced to fully support URLLC use-cases in DECT-2020.

We provide an outlook of DECT-2020 in the next section. Following that, we introduce the use-cases and their requirements. Then, we evaluate the performance of DECT-2020 for selected use-cases. Following that, we discuss the enhancements for support of URLLC services in DECT-2020. Finally, conclusions are provided.

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The DECT-2020 NR physical layer was designed to benefit from recent development in radio technologies and still allow for compatible low-cost implementation. To avoid reliance on local spectrum regulators, the utilized frequency bands are the DECT bands, 1880–1900 MHz band in Europe and Australia, and 1910–1920 MHz band in North America, supporting up to 11 or 5 channels, respectively, with a bandwidth of 1.728 MHz each. Other bands are also supported including those in the sub-GHz region [7]. Coupled with different emitted power allowed for devices, DECT-2020 can cover large territories.

Similar to 3GPP NR, the physical layer builds on cyclic prefix orthogonal frequency division multiplexing (CP-OFDM), allowing for flexible data transmission. To manage communication, DECT-2020 employs a time division duplex (TDD) system, allowing both uplink and downlink transmissions over the same channel. The system also supports both time- and frequency-multiplexing. The physical layer allows operation with different channel bandwidths, symbol lengths, and number of symbols in a slot. This design facilitates harmonious coexistence between classic DECT and DECT-2020 [8].

In the medium access control (MAC) layer, DECT-2020 allows RDs to operate in either fixed termination (FT) or portable termination (PT) mode, or in both modes, see Fig. 1. An RD_{FT} serves as a router for its own cluster, managing local radio resources and allowing RD_{PT} to connect to it. If required, it also forwards data from all connected RD_{PT} to a next-hop RD_{FT} . Both modes utilize the listen before talk (LBT) approach with a binary back-off procedure for uplink transmission. The minimum sensing time for LBT in random access channel (RACH) transmission is designed to be as short as two symbols, allowing for rapid access.

DECT-2020 uses a tree topology. Upon initialization, each RD_{PT} selects RD_{FT} for association based on the received link quality and announced cost, indicating how “good” the path toward the external backend systems is. RDs do not need to establish a data path before initiating communication.

The DECT-2020 MAC layer also offers advanced features, including energy conservation and power control. The former is aimed at increasing the RD’s operational cycle, whereas the latter helps to minimize interference with other devices. Operating over multi-hop topologies, the effect of these functions on the network performance is rather complex [9]. For energy conservation DECT-2020 utilizes the duty-cycle approach. Standardization related to improving energy conservation mechanisms is underway.

USE-CASES AND REQUIREMENTS

In this section, use-cases ranging from mMTC to URLLC and their intermediate forms are introduced, and their key requirements are characterized.

NEAR-URLLC USE-CASES

The modern IoT world is shaped by market needs and driven by use-cases. A top-down approach to designing a communications system should begin with the main stakeholders and their practical needs. At the USWA consortium [6], we have developed a methodology made up of actions

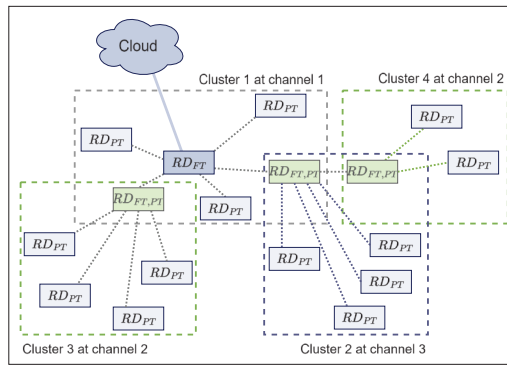


FIGURE 1. Illustration of DECT-2020 network formation.

and activities to bridge technical specifications with real-world use-cases.

Figure 2 shows use-cases identified through a series of contributions and stakeholder interviews. They are grouped according to their performance requirements, based on the framework established in ITU-R M.2410 [4]. The figure categorizes the use-cases into conventional mMTC and URLLC applications, while also highlighting a middle ground: near-URLLC services, which have moderate requirements compared to full URLLC but are still more demanding than mMTC.)

In *industrial factory automation*, the main goal is to replace the currently utilized specialized field buses like EtherCAT and PROFINET. Energy grid applications necessitate the transmission of precise energy Quality-of-Service (QoS) and consumption information from Circuit Monitoring Sensor (CMS) devices. A large class of use-cases, called *off-highway vehicles*, features diverse applications including safety signalling, remote control, and video for applications such as smart agriculture, constructions, and so on. Lastly, specific use-cases, such as *elevator safety signalling* or *audio conference system*, impose stringent latency requirements close to URLLC but may require a limited number of RDs.

USE-CASES REQUIREMENTS

The requirements for the use-cases can be classified by utilizing the Volere methodology into functional, non-functional, and constraints [10]. Functional requirements focus on the core aspects of a product, describing what it needs to do and the rules it must follow. Non-functional requirements are properties that the functions of the system should have, such as performance and usability. Finally, constraints impose restrictions on the solution. They apply to the technology to be used or any other hardware, software, or business practice that must be conformed with.

Table 1 lists the main functional and non-functional requirements for the considered use-cases. These requirements are estimated by USWA industrial consortium members, based on existing standards and practical testbeds. The only constraint currently imposed is compatibility with DECT-2020, as discussed in [8]. From a technical standpoint, the most critical non-functional requirements are those related to system performance, particularly latency, transmission capacity, and packet delivery ratio (PDR). By analyzing the NF requirements further, one may observe that the main factor differentiating use-cases is latency. Many use-cases require latency to fall within the range of tens to

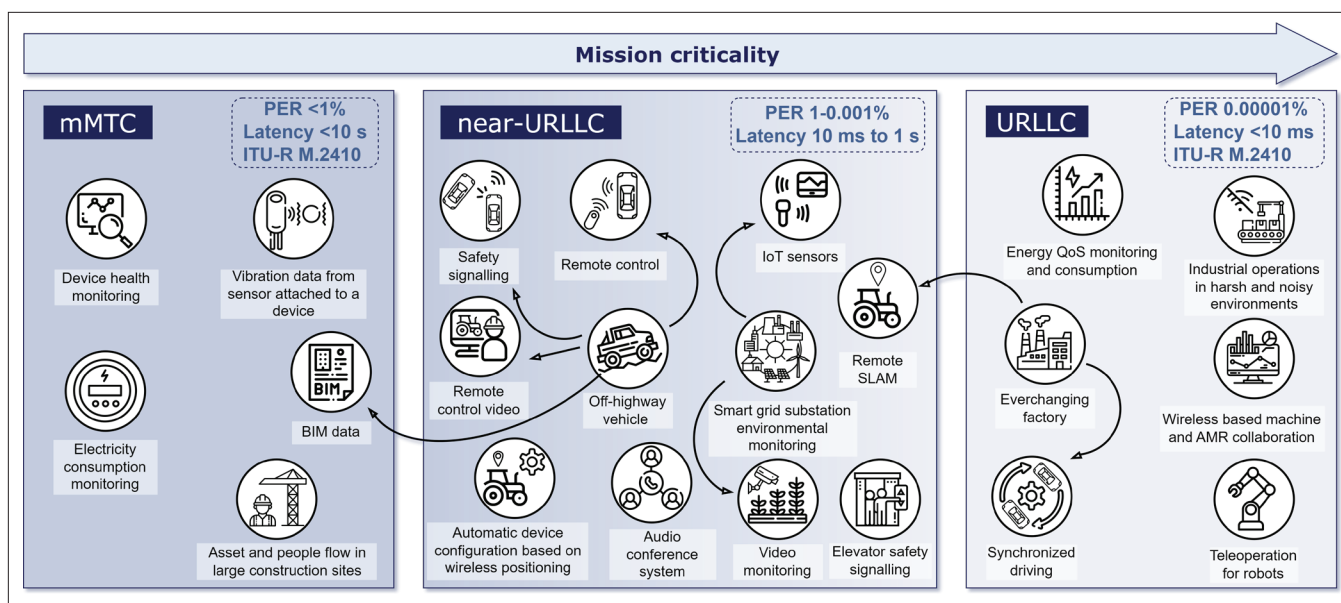


FIGURE 2. Classification of modern mMTC and URLLC use-cases.

a few hundred milliseconds, which is much stricter than mMTC but still far from URLLC.

The use-cases also differ in terms of functional requirements, including the security, communication distance, and deployment density, where latter is expressed as RDs per area, per site, or per facility. While a high level of security is crucial for only a few use-cases, device density is a critical parameter that can influence the NF metrics. As the main medium access method is LBT, in use-cases with high RD density, the self-system interference may affect the system's performance.

As an example, elevator shaft use-case presumes availability of real-time communications between motor and elevator cabin. The communication distance may reach 1000 m for modern skyscrapers implying that multi-hop communication might be required. Depending on the signaling scheme utilized by the vendor, latency and PDR may vary. For example, it is common to detect a failure when more than a single consecutive packet is lost, that is, three packets lost in a row in approximately 50 ms would result in malfunctioning and stoppage of the cabin. Packet loss probability of 10^{-4} would result in one emergency stop in 10 years.

SUPPORT OF NEAR-URLCC USE-CASES

Existing off-the-shelf technologies, like LTE-M and NB-IoT, are designed primarily for conventional mMTC applications. They rely on preamble repetitions and a two-step access procedure, hindering their ability to meet the stringent latency demands of near-URLLC. To mitigate latency, NB-IoT standards were extended in Release 15 to support Early Data Transmission, allowing small packets to be delivered during the random access phase. However, this introduces constraints on preamble ensembles and can potentially reduce coverage. While NR RedCap addresses throughput requirements, it falls short in other areas, such as latency and reliability metrics.

Multi-hop DECT-2020 offers a versatile solution for near-URLLC applications, providing enhanced reliability, reduced latency, and higher bitrates. Its modular architecture facilitates easy deployment

and maintenance of control, monitoring, and protection components. By enabling alternative data transmission routes, the system becomes more resilient to network failure. To meet the stringent latency requirements of near-URLLC services, multi-hop DECT-2020 employs a random access regime. When necessary, a single-hop operational mode can be activated for even lower latency. To ensure high reliability, flexible MCS and HARQ retransmissions are utilized. Operating over a wider bandwidth compared to CIoT technologies, multi-hop DECT-2020 can deliver the required bitrates for near-URLLC applications.

ASSESSMENT OF DECT-2020 CAPABILITIES

In this section, we assess the current state-of-the-art capabilities of DECT-2020 to support near-URLLC applications.

CONSIDERED SCENARIOS

We consider three typical scenarios that resemble near-URLLC use-cases of the previous section, and are characterized by different degrees of interference, topology, and coexistence requirements: isolated URLLC scenario, device-to-device (D2D) single-hop URLLC/mMTC scenario, and multi-hop URLLC/mMTC scenario. Parameters common for all scenarios are shown in Table 2.

The isolated URLLC scenario corresponds to the use of RDs in deployments, where RDs do not experience interference from other networks or from competing mMTC traffic in the same network. We consider an elevator use-case with a sink located at the top of the shaft and RD associated with a moving cabin. Traffic is telemetry with strict latency and reliability requirements. We assess the case of multiple shafts (lanes) located nearby and operating at the same channel.

In the mixed URLLC/mMTC, we address those use-cases, where the network mainly provides mMTC service, but some RDs are configured to operate in the D2D URLLC mode. This may resemble the concert environment, where mMTC is utilized to control specific equipment such as mixers or microphones, while URLLC service provisions voice communica-

Use-case	Functional parameters		Non-functional parameters		
	Device density	Range	PDR	Latency	Rate
mMTC use-cases					
Asset and people flow in large construction sites	>100 RDs/100 m ³	100 m	99 %	1 s	3 Mb/s
Off-highway vehicle – BIM data	100 RDs/site	1000 m	98 %	10 s	2 Mb/s
Electricity consumption monitoring	25000 RDs/km ²	1000 m	99.5 %	1 h	<1 Mb/s
Vibration data from sensor attached to a device	10–100 RDs/facility	200 m	99.5 %	60 s	2.5 Mb/s
Device health monitoring	10-1000 RDs/facility	500 m	99 %	1 s	2.5 Mb/s
Near-URLLC use-cases					
Automatic device configuration based on wireless positioning	>100 RDs/100 m ³	100 m	99 %	100 ms	0.4 Mb/s
Off-highway vehicle – safety signalling	100 RDs/site	1000 m	>99.9999 %	32 ms	5 Mb/s
Off-highway vehicle – remote control	100 RDs/site	1000 m	>99.9999 %	10/300 ms	3 Mb/s
Off-highway vehicle – remote control video	100 RDs/site	1000 m	98 %	300 ms	10 Mb/s
Smart grid substation environmental monitoring – IoT sensors	10-50 RDs/site	100 m	99 %	1 s	<1 Mb/s
Smart grid substation environmental monitoring – video	100 RDs/site	1000 m	99 %	100 ms-1 s	1-20 Mb/s
Audio conference system	10–50 RDs/facility	50 m	> 99.9999 %	< 15 ms	1 Mb/s
Wireless based machine/AMR collaboration in manufacturing	> 1000 RDs/network	100 m	99 %	1 ms	2 Mb/s
Elevator safety signalling	6 RDs/facility	100 m	99 %	16 ms	1 Mb/s
URLLC use-cases					
Energy QoS monitoring and consumption	10–100 RDs/site	1000 m	99 %	1–10 ms	1–10 Mb/s
Industrial operations in harsh and noisy environments	100 RDs/100 m ²	100 m	> 99.999 %	< 1 ms	0.1 Mb/s
Everchanging factory – remote SLAM	200 RDs/30000 m ²	300 m	>99.9999 %	< 30 ms	0.4 Mb/s
Everchanging factory – synchronized driving	200 RDs/30000 m ²	50 m	>99.9999 %	< 10 ms	0.2 Mb/s
Teleoperation for robots	10-100 RDs/facility	10 m	>99.9999 %	< 10 ms	10 Mb/s

TABLE 1. Use-case requirements.

tions. In this scenario, the URLLC RDs experience interference created by the mMTC service.

In the URLLC/mMTC scenario, the URLLC service is provisioned in a multi-hop manner between RDs and a sink. This scenario corresponds to the use-cases with high communication distances, such as energy QoS monitoring, off-highway vehicles. Similar to the previous scenario, there is competing mMTC traffic. For this scenario, we implemented priority queuing for URLLC transmissions.

SIMULATION ENVIRONMENT

The complexity of LBT access operating in a multi-hop communications environment requires the use of a system-level simulation tool for performance assessment. The DECT-2020 system was implemented using the WINTERsim simulation tool which was utilized during the standardization process of DECT-2020 [2].

WINTERsim is a discrete event system-level simulator. For DECT-2020, we used node abstraction as the basic class for RDs, which contains all the node parameters (e.g., position, mobility model, antenna parameters, and power) and serves as a container for L2 interfaces, forwarding, flow tables, signalling engine, application handlers, traffic generators, physical interfaces, and so on. To simulate high-density deployment, we implemented a physical layer abstraction that stores all the bidirectional path loss states between the RDs.

When performing simulations, we followed ITU-R guidelines in M.2412 [5]. For specific environments, such as elevator shafts, we consider an appropriate propagation model [11]. To gather the data, we employed the method of replications. Specifically, we ran simulations 25 times, each time gathering 105 observations in the steady state.

Recall that quality of service (QoS) for mMTC is specified by latency quantile [5]. For URLLC applications not only latency is important but also the PDR. Since we consider near-URLLC services, we utilize both PDR and the 95th percentile of the packet latency as performance indicators.

NUMERICAL RESULTS

In this section, we present and discuss the numerical results. Figure 3a shows the performance of the isolated URLLC scenario for different numbers of communicating pairs representing the elevators in adjacent shafts. We observe that for a single elevator case, the PDR remains almost unaffected at values close to 100 percent whereas the latency is constant at approximately 3 ms. This satisfies the requirement of having latency less than 16 ms in the case of elevators. When more than a single elevator is installed in a shaft interference starts to play a critical role. For two lanes the PDR abruptly drops at high URLLC inter packet times. This is accompanied by an increased 95th latency percentile. Adding one more lane results in even

Parameter	Value
Common parameters	
Carrier frequency	1900 MHz
Number of channels	1
Channel bandwidth	1.728 MHz
RD noise figure	7 dB
RD antenna element gain	0 dBi
Channel access	Random access with LBT
Contention window duration	24–768 symbols
LBT threshold	−75 dBm
Error correction	HARQ with 3 attempts
Sensitivity	−99.7 dBm (TS 103.636-2)
Tx power	10 dBm
Data MCS	#3, #4, #5 for 1, 2, 3 packets
Packet processing time	1 slot
Simulation area	Circle with radius of 170 m
DECT-2020 NR FT	1, height 25 m, outdoors
mMTC RD deployment	Uniformly random
mMTC building penetration loss	20 dB (ITU-R)
mMTC RD indoor/outdoor fraction	80 %/20 %
mMTC channel model, Sink-RD	Urban macro
mMTC Channel model, RD-RD	Urban street canyon
URLLC HARQ	Enabled
URLLC traffic inter packet time	CBR, 4–32 ms
mMTC traffic inter packet time	10–60 s
mMTC traffic load type	Poisson, bursts of 2 packets
Number of mMTC RDs	1000
Isolated URLLC scenario	
URLLC RD deployment	RD pairs along parallel lines
Number of URLLC RDs	2–6 (1–3 pairs)
URLLC RD-Sink distance	100 m
URLLC propagation model	Elevator shaft [11]
D2D URLLC/mMTC scenario	
URLLC RD deployment	Uniformly random in a circle
Number of URLLC RDs	1
URLLC RD-Sink distance	< 50 m
URLLC propagation model	Same as mMTC
Multi-hop URLLC/mMTC scenario	
URLLC RD deployment	Uniformly random in a circle
Number of URLLC RDs	1
URLLC RD-Sink distance	< 100 m, 0–1 hops to sink
URLLC propagation model	Same as mMTC
URLLC message copies sent	3
URLLC/mMTC Tx queuing	Priority-based

TABLE 2. Considered system and evaluation parameters.

worse performance, as the operational regime of the system becomes feasible for rather large inter packet times of 20–35 ms.

Performance of the D2D URLLC/mMTC use-case is shown in Fig. 3b. Here, we observe that mMTC traffic has a limited impact on URLLC performance for considered mMTC inter packet times. Specifically, for URLLC inter packet times of 6 ms or more, the system provides a perfect packet delivery ratio starting from a mMTC inter packet time of 40 ms or longer. The 95th latency percentile in this regime is approximately 3 ms coinciding with that of the single lane case reported in Fig. 3a. These results show that mMTC traffic has less impact on D2D URLLC performance than competing URLLC traffic.

Finally, we assess the performance of the most demanding scenario, where URLLC packets are delivered in a multi-hop way, see Fig. 3c. We compare two cases: when URLLC device is directly connected to the sink, and when the sink is one hop away. Despite the use of priority queuing, we see that even with an inter packet time of 16 ms for URLLC, reliable delivery cannot be ensured even for the lowest considered load of the mMTC service. The latency also increases by more than three times compared to the previous scenario. Even with a direct connection to the sink, the latency increases as both services are implemented in a single network. As reported in [2], 99th latency percentile for NB-IoT and LTE-M in underloaded conditions are around 600–700 ms and 200 ms, respectively. While for DECT-2020 operating in multi-hop way it is around 30 ms, the latency is expected to grow exponentially when traffic load increases. Thus, neither cellular mMTC technologies nor DECT-2020 operating in multi-hop way is capable of supporting near-URLLC applications. Generally, performance degradation is due to two factors: multi-hop impact manifesting itself in latency accumulation when packet needs to go via multiple hops, and additional impact caused by interference at each link.

Summarizing the results, we note that out of the considered scenarios, only that, where multi-hop is involved, is challenging for DECT-2020 to provide the services classified as near-URLLC ones. If one wants to enable them over multi-hop topologies, further standardization efforts are required.

DECT-2020 ENHANCEMENTS FOR URLLC

In this section, we discuss the potential enhancements that may enable strict URLLC services and near-URLLC services over the multi-hop topologies.

PHYSICAL LAYER ENHANCEMENTS

To provide principal feasibility of near-URLLC service provisioning over multi-hop DECT-2020 topologies, utilization of new numerologies with smaller slot time is critical. As noted earlier, DECT-2020 supports multiple numerologies, which provide capabilities to lower the transmission delay and improve the payload data rate over the control data ratio of the transmission, as well as the channel fading performance. With 27 kHz sub-carrier spacing, the slot duration is 416.7 μ s which can be divided into two sub-slots. Increasing the sub-carrier spacing to 54 kHz with 3.456 MHz channel bandwidth doubles the symbols per transmission slot or alternatively reduces the

data transmission time by half when using sub-slots transmission [4], that is, the same amount of data can be transmitted in 208.3 μ s. Using an even higher numerology would repeat this phenomenon and enable further reduction of latency.

NOVEL DATA DISSEMINATION METHODS

Meeting URLLC constraints in wireless mesh networks remains an unresolved challenge. One possible solution is to use network coding, which is typically applied at higher layers in the network stack. It enhances end-to-end communication reliability and reduces the latency caused by multi-hop scenarios using linear algebra over finite fields [12]. With this approach, data of end-to-end nodes are received at intermediate nodes, decoded and used to construct a specific composite data signals (e.g., by XOR operation). These signals are then forwarded toward the destination nodes. By combining multiple received signals the destination node can extract the data intended for it. In the design of the network coding, it is worth noting that URLLC constraints are usually imposed only on the communication between a limited number of nodes within a small area. Therefore, in most cases it is sufficient to apply the network coding to the intra-cluster communication. In some use-cases even a topology agnostic network coding is applicable, avoiding any signaling overhead for topology changes.

As DECT-2020 operates on a TDMA-based system, network coding can optimize scheduling time slots, reducing the total number of transmission between devices and so reducing their energy consumption. For this purpose, methods for constructing this composite signal are needed, focusing on seamless integration into the protocol stack. For an effective implementation, informing RDs about network coding-assisted communication via MAC protocols is necessary.

AI-BASED NETWORK MANAGEMENT

Disabling HARQ may be required to meet the requirements of URLLC. Alternatively, AI-based HARQ methods can reduce the latency of classical HARQ while enhancing network reliability. For instance, Swift HARQ [13] applies a Support Vector Machine (SVM) based algorithm to predict the probability of packet decodability in future retransmissions according to current channel conditions. Based on this, the transmitter reacts proactively by dropping non-decodable packets and reallocating resources for transmissions to RDs with better channel conditions. Similarly, the receiver can reduce latency by omitting unnecessary transmission of negative acknowledgment (NACK) messages during retransmission. Furthermore, the prediction of Swift HARQ enhances system reliability by enabling the transmitter to schedule retransmissions at times when there is a high likelihood of successfully decoding the signal. Initial simulations in [13] indicate that Swift HARQ outperforms classic HARQ mechanisms in terms of two-way latency and block error rate. As part of DECT-2020, RDs using Swift HARQ receive packets and check for errors. If no errors are detected, an acknowledgment (ACK) message is sent. Otherwise, a NACK message is returned with an estimate of how many retransmissions are required to successfully decode the packet. This

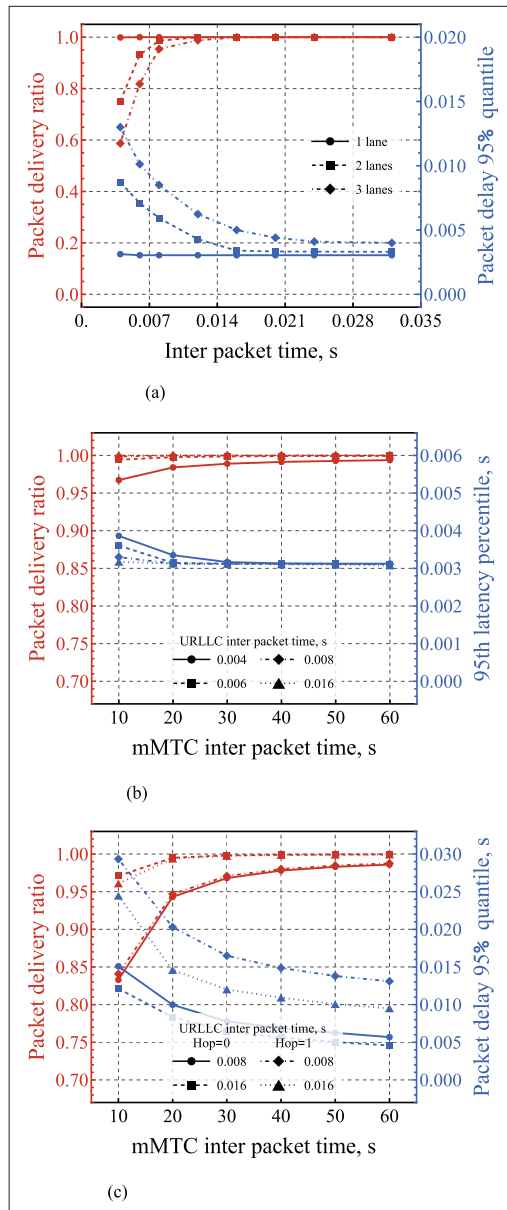


FIGURE 3. Performance of the considered scenarios: a) Isolated URLLC scenario; b) D2D URLLC/mMTC scenario; c) Multi-hop URLLC/mMTC scenario.

will reduce the number of unnecessary retransmissions in the DECT network, thus reducing the transmission delay for communicating RDs.

ADAPTIVE RESOURCE ALLOCATION

Proper allocation of resources is critical for URLLC service. In this context, frequency diversity is essential, yet conventional approaches, such as multi-link operation, face scalability issues. For optimal scheduling, channel-aware resource allocation performed within each cluster was proposed in [14]. A max-min allocation strategy is employed, iteratively optimizing resource allocation based on the SNR. Herein, a quality metric that applies a steeper gradient of a list with the SNRs of each user is the key to a resource allocator. Allocating multiple users per iteration further reduces computational complexity while ensuring adaptability [15]. Channel quality is obtained by the FT node transmitting a sounding sequence and the connected RDs reporting back their channel impulse response. This process ensures that

The DECT-2020 standardization process is currently evolving toward supporting capabilities emphasized above. Once added, DECT-2020 may become the first worldwide non-cellular standard supporting a plethora of new near-URLLC use-cases in a multi-hop way.

RDs entering the cluster are considered while RDs leaving the cluster do not block unused spectrum. Although this represents an additional signaling overhead, the approach presented in [14] outweighs this by efficiently exploiting transmission resources. This method may enhance the performance of networks with dynamic topologies, ensuring reliable communication while maintaining low complexity to meet the stringent latency constraints.

CONCLUSIONS

The aim of this article was to introduce the upcoming near-URLLC scenarios that are envisioned to be supported in a DECT-2020 IoT system, benchmark them, and specify enhancements that need to be implemented to support them.

Our results show that the current implementation of DECT-2020 can provide near-URLLC operation for scenarios where no multi-hop communications are utilized even under interference coming from other D2D URLLC connections or mMTC networks operating at the same frequency. However, multi-hop communication shows a significant decrease in performance. Thus, provisioning of near-URLLC services in DECT-2020 technology requires additional protocol and service delivery enhancements, as outlined in this article.

The DECT-2020 standardization process is currently evolving toward supporting capabilities emphasized above. Once added, DECT-2020 may become the first worldwide non-cellular standard supporting a plethora of new near-URLLC use-cases in a multi-hop way.

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